Know what you want to turn

When you decide to purchase a lathe, it helps to know roughly what you want to make, because no single lathe can do everything you might want to do. Unless you want to work only large scale, it's best to start with a good midi lathe of a style and capacity similar to the one shown at the beginning of this article, and learn the basics on that. A lathe with about 300–405mm faceplate capacity and about 460–600mm between centres capacity allows you to turn a family-sized salad bowl or table lamp as well as most of what I've made over 45 years as a professional woodturner. Working small scale means you waste less material, and when things go wrong it's less dangerous. If you want to turn only long spindles, a small lathe with a bed extension is all you need.

Seek advice from a good turner and indeed your club members. They will be able to offer help with what they know about the lathes they have. Buy wisely and it will save you money in the long run.

I used this lathe for long spindles for several years and it now belongs to my local ACT Woodcraft Guild

What to look for in a lathe

New or second-hand, a lathe should spin the wood at a steady speed with minimal vibration. Consequently a lathe needs to be sturdily built and as heavy as possible with a reasonably powerful motor. The headstock, tailstock, bed and toolrest assembly should be cast-iron to dampen vibration. You don't want anything that flexes or vibrates. My midi lathe weighs 62kg. My stand is constructed of recycled boards 20mm thick screwed to 75mm pine (*Pinus spp.*) legs, and this offers better support than many commercial stands I've encountered for this size lathe.

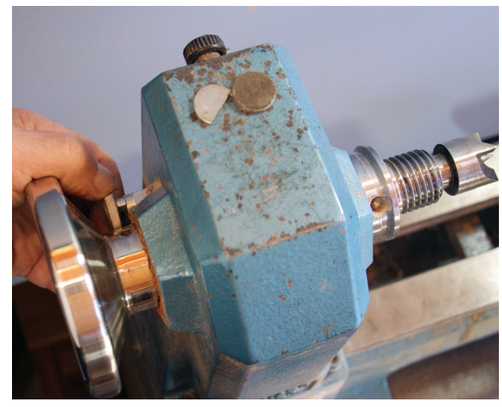
Manufactured bases or stands should be cast-iron or steel at least 5mm thick. Avoid any lathe cast in aluminium or constructed of thin sheet steel. If a lathe is easy to lift, it's probably not worth having… even if it's free.

The headstock

The headstock houses the drive shaft so has to be well constructed. It's essential that there is no play in the bearings. Look for a drive spindle at least 25mm diameter – 30mm is better – mounted in roller bearings. Look for a hollow drive shaft machined to accept a Morse taper spur drive.

Headstocks constructed of thin sheet steel on cheap lathes with the spindle bearings simply clamped to the steel without any additional support cannot possibly support any chuck without rattling, so may be dangerous to use.

A hand-wheel on the outboard (left) side of the headstock is useful as it enables you to bring the lathe to a rapid halt, or rotate a job by hand. You must be able to lock the driveshaft so you can remove a chuck. This can involve a pin, a rod or a wrench. Some lathes come with in built indexing. This is very handy.



The knurled wheel on this lathe locks the drive shaft

The tailstock and rest

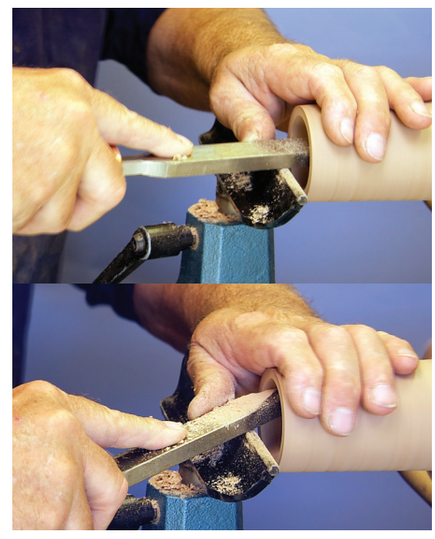
The tailstock and rest assemblies also need to be heavily constructed, with cam action levers that can lock them quickly and easily in position. Using wrenches to lock these components is very tedious by comparison.

The tailstock spindle needs a decent sized wheel to operate smoothly and easily. A tailcentre should wind forward at least 50mm.

The rest must provide absolutely solid support: any play or vibration is magnified by the time it reaches the cutting edge, and this leads to loss of tool control and catches. If you can flex the rest by hand even slightly, choose another lathe.

The toolrest post that slots into the banjo (the bit that sits on the lathe bed) needs to be at least 20mm diameter, and preferably 25mm on all but the smallest lathes. Ideally, the top of a toolrest should slope, like this Robust toolrest with its hardened edge, so a tool can pivot on the top edge. There are many designs, but the rests need to be strong so they don't flex, and allow you to hold the tools well. Flat-topped rests are greatly improved by welding on a 10mm hardened steel rod to the top.

The tailstock and rest assemblies need to be heavily constructed to minimise flex and vibration



The sloping rest on this lathe means a tool can pivot on the top edge

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